

Analyzing the Impedance and Time-Dependence of Signals Propagating Through RG-174 Coaxial Cable

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The signal propagation velocity ($\nu = 65.6\% \pm 0.9\%$ the speed of light) and characteristic impedance ($Z_0 = 50.59 \pm 0.58 \Omega$) were measured for an RG-174 coaxial cable. When terminating the cable with an LC circuit, a relationship between L , C , and the period T was determined as $T_{fit} = (6.643 \pm 0.004)\sqrt{LC}$, although a capacitance-dependence response was observed. Quality factors (Q) were also measured for several LC circuits and compared to their measured resistances. The attenuation coefficient for a 100 MHz signal, $A_1 = 9.777 \pm 0.286$ dB/100 ft was extracted from a square pulse with pulse width $\tau_1 = 54.1$ ns. Limited pulse-spreading was also observed for square waves, consistent with modeled behavior.

I. INTRODUCTION

Since Oliver Heaviside's invention of the coaxial cable in the 1880's, coaxial cables have been especially useful for transmitting high-frequency signals between electrical components for use in telecommunications and detection-based experiments. Understanding and characterizing how these signals propagate through these cables, including how they can distort and attenuated based on pulse shape and frequency, is especially important for proper use in these applications.

The Radio Guide (RG) series of cable is a particularly common coaxial cable that has been well-studied by researchers and the US military in the 1960's and 1970's [1, 2]. While RG cables come in various diameters for different use cases, the RG-174 is a miniature, lightweight series of cable that is still used today by some for radio- or internet-based applications [3]. Compared to other RG cables with larger diameters, RG-174 is more prone to signal attenuation and distortion, which makes it an ideal option to study further.

A major factor for signal propagation through coaxial cables involves how the cable is terminated, or the properties of the electrical equipment it is connected to. Most cables are designed with a particular characteristic impedance Z_0 in mind, so that connections between cables and components can match in impedance and mitigate signal reflection. However, time-dependent effects can dominate when cables are terminated with capacitive or inductive loads, and very short signals can have unexpectedly high attenuation due to skin-effect losses, where signals do not have time to distribute evenly across conductive surfaces [4].

For an ideal transmission line, any signal with a voltage $V(z, t)$ must follow the differential equation

$$\frac{\partial^2 V}{\partial z^2} = LC \frac{\partial^2 V}{\partial t^2}, \quad (1)$$

where L and C are the distributed inductance and capacitance of the transmission line [5]. Solving for V and I and then applying Ohm's Law yields a characteristic

impedance Z_0 of the transmission line, given by

$$Z_0 = \sqrt{\frac{L}{C}}. \quad (2)$$

If we then terminate this transmission line with any set of linear circuit elements with impedance Z , we can compare the voltage of the incident wave V_{in} with the outgoing V_{out} to find a reflection coefficient Γ , defined as

$$\Gamma \equiv \frac{V_{in}}{V_{out}} = \frac{Z - Z_0}{Z + Z_0} \quad (3)$$

[6]. It is important to keep in mind that while impedance is often defined or calculated using a real number (like resistance), it is more fundamentally a phasor that is frequency-dependent. Thus, the period T or pulse width τ of signals can have a major effect on their shape and attenuation as they travel down a transmission line or cable.

If a transmission line is terminated with an LC load, then the period of oscillations for a given signal are given by $T = 2\pi\sqrt{LC}$, and the quality factor Q of those oscillations are modeled by

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}, \quad (4)$$

where R is the effective resistance of the circuit.

II. EXPERIMENT AND APPARATUS

Figure 1 shows a diagram for the two setups used to collect data. In both cases, a function generator was used to generate a variety of signals, including short, gaussian pulses with widths from 10 – 25 ns and longer, square pulses with widths from 50 ns – 5 μ s. These signals were initially sent to an oscilloscope on a short cable to establish a starting V_{in} .

In Setup A, this short cable was connected via a T-junction to longer coils of cable (varying in lengths of 100 – 800 ft) which were then terminated with some circuit consisting of linear circuit elements (resistors, capacitors, and inductors) wired in series. These long cables

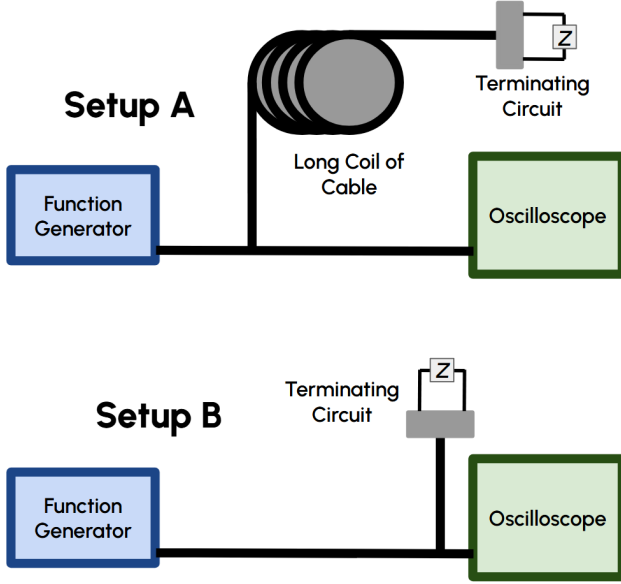


FIG. 1. Schematic diagrams of the setups used in this experiment. For the majority of data collected, Setup A was used. A function generator sends short pulses or square waves (With pulse length $\tau \sim 10 - 100$ ns) through an RG-174 coaxial cable. Part of this cable is sent directly to an oscilloscope, while another branch leads to a much longer cable (length $l \sim 100 - 800$ ft). This cable is then terminated by either shorting the inner and outer conductors, leaving them open, or using some combination of linear circuit elements with impedance Z . Setup B is very similar, except the long coil is omitted.

allowed short pulses to be separated enough to independently measure V_{out} of the reflected signals. In Setup B, another short cable was used instead of a long cable, also terminated with varying circuit elements. V_{in} and V_{out} now arrive at the oscilloscope too close together to be easily distinguished, but attenuation effects of the cable are mitigated to allow for easier measurement of certain time-dependent signals, such as ringing produced by a terminating LC circuit.

All cables used in this experiment were RG-174 cables that were previously cut to known lengths. Thus, any material properties that were measured for these cables (such as Z_0) can be compared to their reported values in other references, such as the MIL-DTL-17 standards or the manufacturer datasheet [2, 7].

Between some figures, the same particular capacitors (with capacitance C_1 , C_2 , and C_3) and inductors (L_1 , L_2 , and L_3) were used repeatedly. These circuit elements were measured using an LCR meter which was accurate to beyond five significant figures. Their values in the report will be reported with three or four digits, but all fits used their more exact values. This LCR meter was also used to determine the internal resistance of the inductors (labeled Q_1 , Q_2 , and Q_3 , respectively).

III. RESULTS

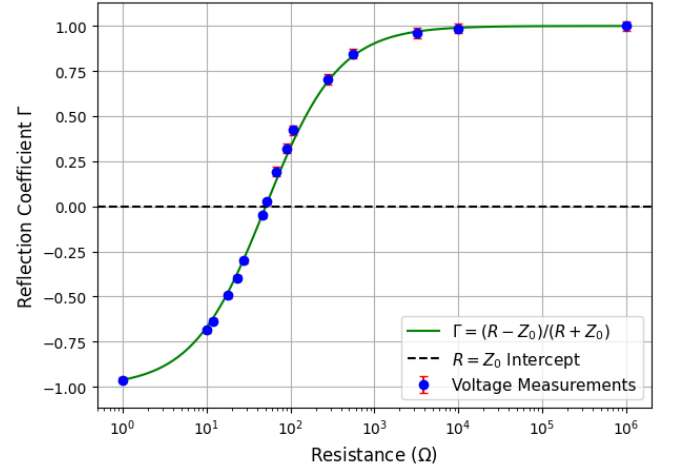


FIG. 2. Measured resistor values plotted with their corresponding reflection coefficient $\Gamma = V_{in}/V_{out}$. The green curve is a least-squares fit for the function $\Gamma = (R - Z_0)/(R + Z_0)$, where R is the terminating resistance and Z_0 is the characteristic impedance of the RG-174 cable (the free parameter). The intercept of the model at $\Gamma = 0$ gives us this characteristic impedance of $Z_0 = 50.59 \pm 0.58 \Omega$.

Figure 2 produces a plot of the reflection coefficient Γ for various terminating resistors using Setup A in Figure 1. To produce the plot, the peak-to-peak voltage of an incoming pulse V_{in} was measured, and then a corresponding V_{out} was measured for each resistor. To set the bounds for Γ , these voltage values were offset so that an open termination corresponds to $\Gamma_{open} = 1$ and a short corresponds to $\Gamma_{closed} = -1$. With these bounds, each resistance could be associated with a measured Γ , and a fit could be applied to the data to determine Z_0 , the impedance of the RG-174 cable (which is also represented by the intersection of $\Gamma = 0$ with the fit). The fit found that $Z_0 = 50.59 \pm 0.58 \Omega$, with $\chi^2 = 1.29$. Uncertainties were driven by the resolution of the oscilloscope, with negative reflections ($\Gamma < 0$) having a resolution of 20 mV and positive reflections ($\Gamma > 0$) having a resolution of 40 mV due to the limitations in triggering signals.

Using Setup A, other terminations (consisting of resistors, capacitors, and inductors in series) were used to qualitatively measure the shape of reflections. With just a capacitor or resistor terminating the long coil of cable, the pulse would return with a positive lobe followed by a negative lobe (capacitor), or a negative lobe followed by a positive lobe (inductor). This follows the expectations from our initial differential equation. Adding resistors in series with these connections damped the observed pulses.

A modified form of Setup A (with the terminating circuit end of the long cable being plugged back into the oscilloscope) was used to obtain the speed of signal propagation in the cable. With a cable length $l = 503$ ft, the

time for a ~ 500 ns square wave to return to the oscilloscope was measured using the time difference between the initial rise of V_{in} and V_{out} . With this, the signal propagation velocity ν is approximated as $\nu = 65.6\% \pm 0.9\%$ the speed of light.

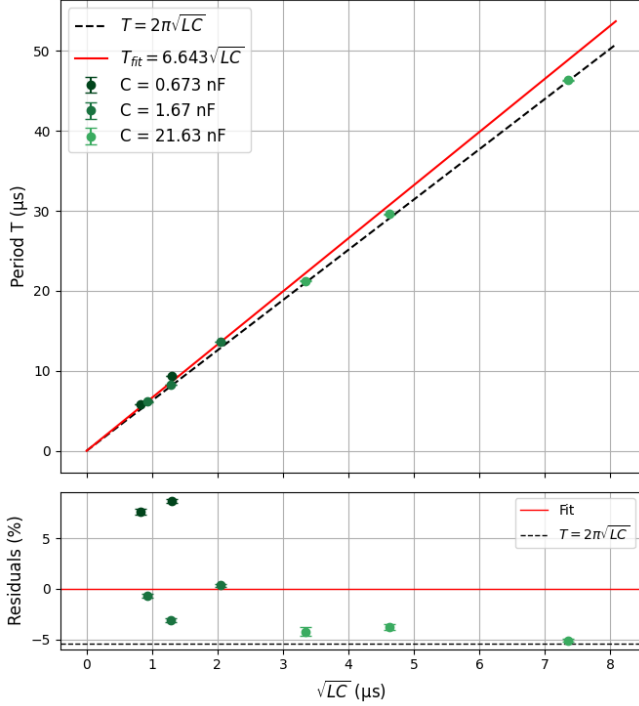


FIG. 3. Top: A plot (and residuals) of $\sqrt{LC}$ for several inductor-capacitor pairs against their measured period when terminating the cable. The theoretical model, $T_{model} = 2\pi\sqrt{LC}$, is plotted along the dotted black line. The best-fit line, $T_{fit} = 6.643\sqrt{LC}$, is plotted in solid red. Each data point is color-coded based on the capacitor that was used. The residuals plots the data relative to the line of best fit. The C_2 data appears to hover around the line of best fit, whereas the C_1 data is well above and the C_3 data is well below, closer to the model.

The top of plot Figure 3 maps the $\sqrt{LC}$ to the measured period for several inductor-capacitor pairs terminating the cable from Setup B. Once plotted, a linear fit was applied to the data, obtaining a line, T_{fit} , with a slope of 6.643 ± 0.004 (plotted in solid red). The model, $T_{model} = 2\pi\sqrt{LC}$, has also been plotted in dotted black. To more easily see the spread of the data, a residuals plot for T_{fit} has also been produced below the main plot. From the residuals, it appears that the data may be grouped by the capacitance used. For the points where $C_3 = 21.63$ nF, the data appears to align closer to T_{model} than T_{fit} . However, for the much smaller $C_1 = 0.673$ nF, measured data is well above T_{model} and T_{fit} .

The quality factor Q for LC circuits in Setup B (see Figure 1) were also measured by plotting the amplitude of subsequent peaks over time, as seen in Figure 4. Only a capacitance of $C_2 = 1.67$ nF was used to generate fits,

since its ringing effects were the most well-defined. For all three inductors, the pulse width was tuned to magnify the amplitude of the ringing as much as possible, and then the peak voltage value was recorded for eight peaks. Since later peaks were more spread out and less reliable indicators of attenuation, errors were calculated using the formula $R\sqrt{n+1}$, where R is the resolution of the scope and n is the number of the current peak-to-peak pulse width being measured. For each inductor, a quality factor could be extracted from the fit, yielding values of $Q_1 = 12.26 \pm 0.18$, $Q_2 = 14.81 \pm 0.14$, and $Q_3 = 15.80 \pm 0.21$ (corresponding to L_1 , L_2 , and L_3 , respectively). Then, for each LC circuit, the LCR meter was used to measure their internal resistances, which came out to $R_1 = 7.33 \Omega$, $R_2 = 10.63 \Omega$, and $R_3 = 13.08 \Omega$ respectively.

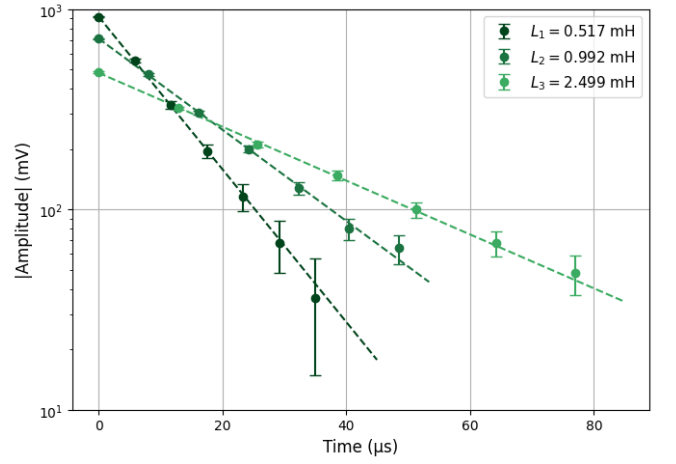


FIG. 4. Subsequent peak amplitudes of a ringing LC circuit terminating an RG-174 cable. A consistent $C = 1.67$ nF was used across these three inductors. Since later pulses were weaker in amplitude and more spread out, their amplitudes have much more uncertainty. Using this decay, quality factors of $Q_1 = 12.26 \pm 0.18$, $Q_2 = 14.81 \pm 0.14$, and $Q_3 = 15.80 \pm 0.21$ were determined from the best-fit lines.

Figure 5 was generated using Setup A with an open end. Square pulses with initial widths $\tau_1 = 54.1 \pm 0.4$ ns and $\tau_2 = 107.0 \pm 0.8$ ns were sent down long lengths of cable and reflected back to the oscilloscope. These reflected pulses could then have their peak-to-peak voltages measured and compared to the initial square pulse. Lines of best fit were then plotted for both sets of data, representing a particular attenuation coefficient α (measured in dB/100 ft) for that pulse width τ . This could then be scaled to a representative 100 MHz square wave (pulse width $\tau = 10$ ns) to determine A , the general attenuation coefficient. Each of the two data sets were considered independently when fitting to A . The smaller pulse width τ_1 measured $A_1 = 9.777 \pm 0.286$ dB/100 ft, and τ_2 measured $A_2 = 11.713 \pm 0.333$ dB/100 ft. When determining which pulse widths to use, it was observed that the function generator struggled to produce square

waves that were any shorter (as they began to warp and appear more Gaussian), and pulses that were longer appeared to have some capacitive response that made the peaks of the square waves drift upwards across the pulse.

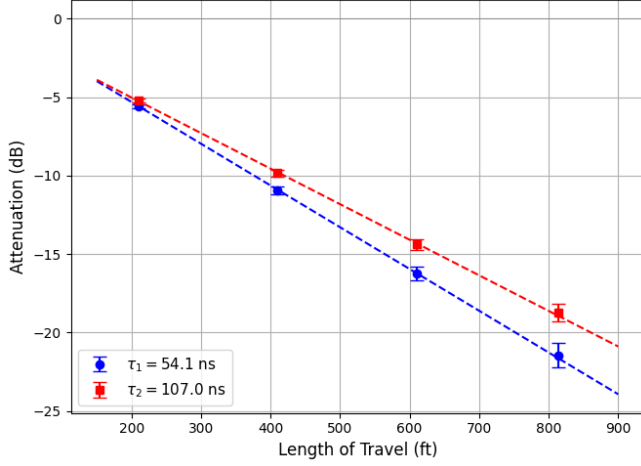


FIG. 5. The attenuation of two square pulses as a function of their total length traveled. The blue data represents an initial pulse width of $\tau_1 = 54.1 \pm 0.4$ ns and the red a pulse width of $\tau_2 = 107.0 \pm 0.8$ ns. Linear fits were applied to these two sets of data to determine their particular attenuation coefficients. Then, both of these were scaled to independently determine A , the attenuation coefficient for a representative 100 MHz square wave (in dB/100 ft). For this data, $A_1 = 9.777 \pm 0.286$ dB/100 ft, and $A_2 = 11.713 \pm 0.333$ dB/100 ft.

Finally, using the A_1 attenuation coefficient calculated from Figure 5, a plot analogous to a figure found in UCRL3307, Section CC2 (Figure 5) was produced [1]. In Figure 6, the y-axis represents the stretching of a square pulse (where T^1 is the initial pulse width and T_1 is the final pulse width). The x-axis is the function $Al\sqrt{10^{-9}/T}$ (where l is the travel length of the signal). For function values less than 1,000, CC2 predicts T_1/T should be between 1 and 1.005. To get further right on the x-axis where T_1/T grows to at least 1.5, a function value of $\sim 3,000$ would be needed, which is beyond the capabilities of the oscilloscope.

The data plotted in figure 2 appears to adhere to the model quite well. A lack of small resistors ($< 10 \Omega$) made it more difficult to fill the left side of the graph, but the data is generally well-spaced and suitable for applying a fit. The measured impedance $Z_0 = 50.59 \pm 0.58 \Omega$ is in agreement with the manufacturer's datasheet of $50 \pm 2 \Omega$ [7]. When measuring, there did appear to be some capacitive or otherwise non-ideal response in the reflected signal (especially apparent near the $\Gamma = 0$ point), but measuring the reflected signal at a consistent time

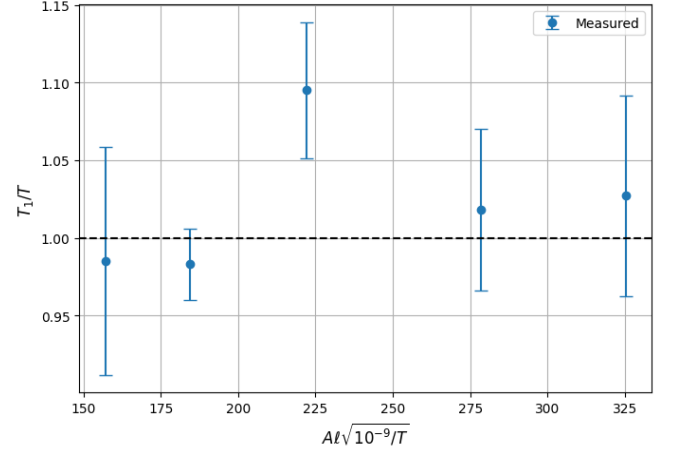


FIG. 6. A plot of T_1/T , the spreading of an initial square wave signal, as a function of the attenuation coefficient A , the length of the cable l , and the width of the input square wave T (T_1 is the width of the ending signal). This plot is a direct comparison of the model in Figure 5 of UCRL3307, Section CC2 [1]. Since all of the data falls on x-axis values well below 1000, the model predicts that T_1/T should be very nearly 1 for all the plotted data.

across the data appear to mitigate its impact. Terminating the circuit with a roughly 50Ω resistor in Setup B also produced the expected results of minimizing reflections at the oscilloscope. This characteristic impedance Z_0 provides a good baseline for understanding the cable at lower-frequencies.

The measured signal propagation, $\nu = 65.6\% \pm 0.9\%$ the speed of light, is also in agreement with the manufacturer data sheet report of 66% and the 65.9% found in CC2 [1, 7]. When measuring the start or end of signals with the oscilloscope, it can be difficult to determine the error associated with those measurements, since the resolution of the oscilloscope is often greater than the stability of the signal (especially for a V_{out} that has traveled a significant difference). Capturing many traces was useful for determining such bounds.

The data in Figure 3 for ringing periods appears to be in loose agreement with the model, but the slope of the fit line is significantly different from the predicted 2π and the small uncertainties of the data disagree with this fit. The residuals plot is especially insightful for interpreting the data. The largest C_3 capacitors trend much closer to the model line while the smaller capacitors trend above it, which could be an indication that the internal capacitance of the cable is having a significant effect. Indeed, if we use the reference sheet value of 0.101 nF/m, we find that for our five foot cable, we expect an internal capacitance of roughly 0.154 nF, which is on the order of magnitude of $C_1 = 0.673$ nF and $C_2 = 1.67$ nF. With more data across many capacitors, it may be possible to add this internal capacitance as a free parameter to a fit for the period.

For the quality factors extracted from Figure 4, it

¹ T is used here instead of τ for disambiguation with the data from Figure 5

should be noted that the error scaling used ($\sqrt{n+1}$) was chosen somewhat arbitrarily, but some error scaling was necessary since determining uncertainty from just the oscilloscope readings themselves was difficult. The first few peaks of each data set were especially sharp and easy to extract, but when the amplitude fell below 10% of the initial, the data became unreliable. Unlike extracting the period from these LC pairs, determining Q was much more sensitive to the capacitance chosen, as capacitors other than C_2 were either prone to ringing which decayed too quickly or became too noisy, likely due to the internal capacitance of the cable (as previously mentioned).

The quality factors for each inductor are all below their measured resistances, although the largest difference was the for smaller inductor ($Q_1 = 12.26 \pm 0.18$ compared to $R_1 = 7.33 \Omega$). Given Equation 4, we expect that the quality factor is representative of the total effective resistance in the circuit, including the inductor, capacitor, cable, and connections between the function generator and oscilloscope. This difference is magnified when the internal resistance of the inductor is smallest, as with L_1 .

It is clear that the two fits are in disagreement in Figure 5, largely due to the limitations in generating a clean square pulse in the RG-174 cable. In CC2, Figure 7 shows that after around 220 ft, the V_{out} pulse shape is already difficult to define, which was the starting set of measurements used for the attenuation coefficient data [1]. The reference value given for A (at 100 MHz) is given as $A = 9.0$ dB/100 ft, which is somewhat close to the value measured with a pulse width $\tau_1 = 54.1$ ns, which was $A_1 = 9.777 \pm 0.286$ dB/100 ft. However, as indicated by the variation in A_2 and the pulse-shape issues described, neither of these measurements are likely to be accurate measures of the true attenuation.

From Nahman, we expect that attenuation follows $\alpha = a\sqrt{\frac{2\pi}{\tau}} + b$, where a represents the frequency-dependent losses that stem from the skin effect and b represents the constant (DC) losses in the cable. [4]. Since the data collected here used pulses $\gg 1$ ns, these skin-effect losses dominate, so we can effectively ignore the b term. Using

this simplification, the manufacturer's datasheet provides a value of $A = 8.5$ dB/100 ft which is notably lower than the reported value in CC2 [7].

Due to the limitations of the oscilloscope, the plot in Figure 6 is in agreement with the expected limited pulse-spreading, but is unable to test parts of the model where T_1/T deviates from 1. Making the cable any longer would only further increase the uncertainty in the measurements (due to too much attenuation) which would make it impossible to extract any interesting features, and the function generator used simply could not generate short enough pulses to observe a significant pulse-spreading effect.

IV. CONCLUSION

Across all of the figures, the effects of characteristic impedance Z_0 , signal propagation speed ν , signal attenuation, and signal spreading were observed and measured with varying degrees of accuracy. Since the LCR meter was able to determine resistances, capacitances, and inductances with a high degree of accuracy, the fits used to determine Z_0 and Q were quite accurate and largely limited by the resolution of the oscilloscope (outside of the later peaks shown in Figure 4). The error for measuring the propagation speed ν was also only limited by the resolution of the oscilloscope.

Other, more time-dependent measurements, such as those used in Figures 3, 5, and 6, had other major sources of error that broadened their statistical error or introduced other systematic uncertainties that were difficult to quantify. In the case of measuring the period T in the LC-terminating cable, the capacitance of the cable itself or other parasitic sources clearly had an impact on the measured period values, seen by the grouping of the data by capacitor. The square-wave attenuation data was also limited by the misshapeness of the signals produced by the function generator, making both measured A_1 and A_2 values unreliable, even if A_1 loosely corresponded with other reference values. The limitations in pulse length also impacted the ability to measure signal spreading in Figure 6.

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